In many ways, you can think of an identity provider as a traditional web app that has only account management pages. If you want to take on building your own identity provider, ASP.NET Core Identity, described in chapter 23, provides a good basis for the user management side. Adding IdentityServer or OpenIddict gives you the ability to generate tokens for other services, using the OpenID Connect standard, for maximum interoperability with other services.

25.1.3 OpenID Connect and OAuth 2.0

OpenID Connect (OIDC) (http://openid.net/connect) is an authentication protocol built on top of the OAuth 2.0 (https://oauth.net/2) specification. It's designed to facilitate the kind of approaches described in section 25.1.2, where you want to leave the responsibility of storing user credentials to someone else (an identity provider). It provides an answer to the question "Which user sent this request?" without your having to manage the user yourself.

NOTE It isn't strictly necessary to understand these protocols to add authentication to your APIs, but I think it's best to have a basic understanding of them so that you understand where your APIs fit into the security landscape. If you want to learn more about OpenID Connect, *OpenID Connect in Action*, by Prabath Siriwardena (Manning, 2023), provides lots more details.

Open ID Connect is built on top of the OAuth 2.0 protocol, so it helps to understand that protocol a little first. OAuth 2.0 is an *authorization* protocol. It allows a user to delegate access of a resource to a different service in a controlled manner without revealing any additional details, such as your identity or any other information.